The empirical recurrence for OEIS A268881, proved by a counting transfer matrix

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A268881 counts arrays over $\{0, \ldots, 1\}$ subject to a condition on the array as a whole rather than cell by cell: the number of adjacent pairs with a stated property must be exactly one. A transfer matrix still applies once the running count is carried in the state; and because a count of two or more can never come back, those states may be dropped outright, which is what keeps the digraph sparse. That makes $a(n)$ a walk count on $S = 16$ vertices, and the empirical recurrence of order 4 contributed by R. H. Hardin then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A268881 is "Number of n X 3 binary arrays with some element plus some horizontally or antidiagonally adjacent neighbor totalling two exactly once.". It has offset 1 and begins

$$2, 14, 84, 462, 2418, 12252, 60666, 295230, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 10*a(n-1) - 31*a(n-2) + 30*a(n-3) - 9*a(n-4).

As of the "Last modified" line on the live entry (Jan 15 2019 16:11:26, revision 7) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \; = \; 10\,a(n-1) - 31\,a(n-2) + 30\,a(n-3) - 9\,a(n-4) \tag{1}$$

for all $n > 4$.

2 A global count, and how to carry it

The arrays have 3 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u. Call two cells *adjacent* when they differ by one of the offsets

$$\mathcal{N} = \{(0, 1), \ (1, -1)\},$$

written as (row shift, column shift), and call an adjacent pair $\{p, q\}$ *marked* when

$$x_p + x_q = 2.$$

Each unordered adjacency is represented by exactly one offset, all with a nonnegative row shift, so every pair is named once and only once: the pairs inside a row are seen when that row is laid down, and the pairs joining two rows when the later of the two is. The entry's requirement is

$$\#\{\text{marked pairs in the whole array}\} \; = 1. \tag{2}$$

This is not a condition on any bounded window, so no transfer matrix over rows alone can express it. It becomes one as soon as the running total is part of the state. Let

$$\mathcal{V} = \big\{(r,c): \; r \in \{0,\dots,1\}^3, \; c \in \{0,1\}\big\}, \qquad |\mathcal{V}| = S = 16,$$

and for rows r, s write $\beta(r,s)$ for the number of marked pairs joining them plus the number of marked pairs inside s. Declare $(r,c) \to (s, c+\beta(r,s))$ an edge whenever $c + \beta(r,s) \le 1$, and no edge otherwise.

Remark 1. Dropping the states with $c \ge 2$ is not an approximation. Once the total has reached two it can only grow, so no array satisfying (2) passes through such a state. Keeping them — as a third, absorbing value of c — would make every pair of rows an edge and the digraph dense, with $(1+1)^{2\cdot 3}$ edges instead of the sparse set used here.

Lemma 2. *Let* $\mathbf{v}$ *mark the vertices* (r,c) *whose count* c *equals the number of marked pairs inside* r *alone, and* $\mathbf{u}$ *the vertices with* $c = 1$*. Then for* $L \ge 1$ *the arrays of* L *rows satisfying* (2) *are exactly the walks of length* $L-1$ *from a vertex marked by* $\mathbf{v}$ *to one marked by* $\mathbf{u}$*, so*

$$a(n) \;=\; \mathbf{v}^\top N^{\,L-1}\mathbf{u}, \qquad L = n.$$

Proof. A walk $(r^{(1)}, c_1) \to \cdots \to (r^{(L)}, c_L)$ determines the array $r^{(1)}, \dots, r^{(L)}$, and conversely each array determines at most one walk, since the starting count is fixed by $\mathbf{v}$ and each step's count is determined. By induction c_t is the number of marked pairs in the first t rows: it holds at $t = 1$ by the definition of $\mathbf{v}$, and the step from t to $t+1$ adds exactly the pairs joining rows t and $t+1$ together with those inside row $t+1$, which are precisely the marked pairs of the first $t+1$ rows not already counted — each unordered pair being named by exactly one offset. The walk exists (no edge is missing) if and only if the running total never exceeds 1, which for a nondecreasing total is the same as $c_L \le 1$; and $\mathbf{u}$ imposes the entry's requirement on c_L. Note the state already carries one row, so L rows correspond to a walk of length $L-1$. $\square$

3 The criterion, and the computation

Let $q(t) = t^4 - \sum_i c_i\, t^{4-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 3. *With* n_0 *the least index for which* $L \ge 1$ *and* $h_j = G^j N^o \mathbf{u}$*,* $o = 1n_0 + 0 - 1$*,*

$$a(n_0+j) - \sum_i c_i\, a(n_0+j-i) \;=\; \mathbf{v}^\top G^{\,j-4} w, \qquad w = h_4 - \sum_i c_i h_{4-i},$$

for $j \ge 4$*. Hence* (1) *holds from that point on if and only if* $\mathbf{v}^\top G^m w = 0$ *for every* $m \ge 0$*, and it is enough to check* $m = 0, 1, \dots, S-1$*.*

Proof. Substitute Lemma 2 and factor. By Cayley–Hamilton G^S is an integer combination of $I, G, \dots, G^{S-1}$, so every later value is the same combination of earlier ones. $\square$

Theorem 4. *The recurrence* (1) *holds for every* $n > 4$*, which is the statement of Section 1.*

Proof. The digraph was built directly from the definition of a marked pair. The vector w was formed by 4 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 4+1$ onwards, and the run continued until $S+1$ consecutive zeros had been seen, which by Lemma 3 settles every larger m. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the walk counts of Lemma 2 were compared against the entry's DATA at every one of the 22 published indices, with no shift fitted: the number of rows is $L = n$, read off the entry's own name, and the offset says which index the first published term carries. The values agree exactly. This is where an error in the pair bookkeeping would show: counting an unordered pair twice, or missing the pairs inside the first row, changes the counts immediately. Indeed a first version of the computation placed the walk at length $L-2$ rather than $L-1$ — the state carries a row, not a boundary — and every value came out one index late, which the DATA comparison rejected.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 3 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

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